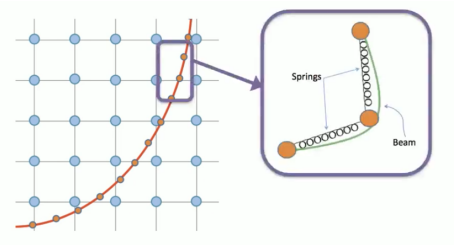
# MOTIVATION FOR THE IMMERSED BOUNDARY METHOD

Classical Peskin formulation, tradeoffs, and method selection

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Fluid Dynamics and IBM Theory • Module 1




Schematic after the Wikimedia Commons figure *ImmersedBoundary grid* (Immersed boundary method article).

Claude (Anthropic) was used to assist with formatting, organization, and clarification of the instructor's original notes.

# WHAT IS THE IMMERSED BOUNDARY METHOD?

The immersed boundary (IB) method is a computational framework for fluid–structure interaction in which

- fluid momentum is described on a fixed Eulerian grid;
- structure position and mechanics are described with moving Lagrangian coordinates;
- integral transforms transfer force to the fluid and velocity back to the structure.

structure force $\xrightarrow{S}$ fluid force,

fluid velocity $\xrightarrow{J}$ structure motion.

The spreading operator $S$ and the interpolation operator $J$ are defined below: continuum forms on the “Continuum force and velocity coupling” slide, discrete forms $S_h$, $J_h$ on the slide after it.

Introduced by Charles S. Peskin for cardiac fluid dynamics, the framework now supports many thin-boundary and volumetric FSI models.

# WHY USE AN IMMERSED FORMULATION?

**Body-fitted fluid meshes**

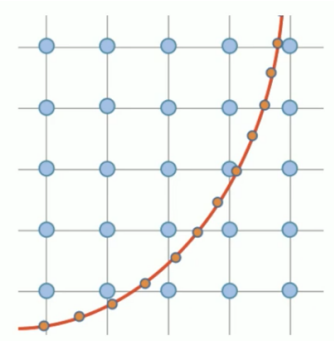
- conform to the moving fluid–structure interface;
- can resolve interfacial flow and stress sharply;
- may require mesh smoothing or remeshing when motion, deformation, or contact degrades mesh quality.

**The immersed alternative**

- keeps the Eulerian fluid grid fixed;
- moves a nonconforming Lagrangian representation of the structure through that grid;
- couples the two descriptions by transferring force and velocity.

> Geometric flexibility comes from the nonconforming fluid and structure descriptions. The Lagrangian structure is still tracked explicitly.

# THE CLASSICAL PESKIN IB METHOD



Blue: fixed Eulerian grid. Orange: moving Lagrangian points. Schematic after the Wikimedia Commons figure *ImmersedBoundary grid* (Immersed boundary method article).

- Charles S. Peskin introduced the method to study flow around heart valves in the 1970s.
- The classical formulation couples an elastic boundary to a viscous incompressible fluid.
- Velocity, pressure, and momentum use an Eulerian description. Structure position and elasticity use a Lagrangian description.
- Later IB extensions treat volumetric solids, active materials, prescribed motion, and constrained rigid bodies.

Applications include cardiac flow, cilia and flagella, animal locomotion, flexible beams, and membranes.

Peskin, *J. Comput. Phys.* 10 (1972) and 25 (1977); Peskin, *Acta Numerica* 11 (2002); Griffith and Patankar, *Annu. Rev. Fluid Mech.* 52 (2020).

# REPRESENTATIVE APPLICATIONS

## Biological fluid mechanics

- heart valves, chambers, and vascular flow;
- cilia- and flagella-driven transport;
- swimming by fish, jellyfish, and microorganisms;
- flexible airways and tissue–fluid interaction.

## Engineering and soft matter

- flapping flags, membranes, and beams;
- flow-induced vibration and mixing;
- bio-inspired propulsion and soft robotics;
- compliant or porous surfaces, with extensions tailored to the required physics.

> The common feature is not a single Reynolds-number regime; it is a moving or deforming interface that benefits from a nonconforming fluid grid.

# WHAT THE CLASSICAL IB METHOD IS—AND IS NOT

**It is**

- a force-based FSI formulation;
- an Eulerian–Lagrangian coupling framework;
- a method whose standard discretization uses a regularized delta kernel;
- compatible with large structural motion.

**It is not**

- a body-fitted or intrinsically sharp-interface method;
- a method that discards elasticity or stress;
- automatically topology changing—the structural representation must model any change;
- restricted to one constitutive law.

> At the continuum level no slip is pointwise. “Weak enforcement” describes the regularized discrete interaction, not the singular continuum equations.

# CONTINUUM FORCE AND VELOCITY COUPLING

Let $q \in \mathcal{U}$ label a material point and let $\boldsymbol{X}(q,t)$ be its position in the fluid domain $\Omega$.

**Spread structural force to the fluid**

$$\boldsymbol{f}(\boldsymbol{x},t) = \int_{\mathcal{U}} \boldsymbol{F}(q,t)\,\delta(\boldsymbol{x} - \boldsymbol{X}(q,t))\; dq.$$

$\boldsymbol{F}$ is a force density per unit material measure. The Dirac distribution places that force at the current structure location.

**Interpolate fluid velocity to the structure**

$$\boldsymbol{U}(q,t) = \frac{\partial \boldsymbol{X}}{\partial t} = \int_{\Omega} \boldsymbol{u}(\boldsymbol{x},t)\,\delta(\boldsymbol{x} - \boldsymbol{X}(q,t))\; d\boldsymbol{x}.$$

For continuous $\boldsymbol{u}$, the Dirac identity gives

$$\frac{\partial \boldsymbol{X}}{\partial t}(q,t) = \boldsymbol{u}(\boldsymbol{X}(q,t),t)\,.$$

> At the continuum level, the kinematic equation imposes pointwise no slip on an impermeable material interface. It is not merely an averaged boundary condition.

In operator form, $\boldsymbol{f} = S[\boldsymbol{X}]\,\boldsymbol{F}$ and $\partial_t \boldsymbol{X} = J[\boldsymbol{X}]\,\boldsymbol{u}$ with $J = S^*$, the $L^2$ adjoint of $S$. Adapted from Griffith and Patankar (2020), §2.1.1, Eqs. 1–4, and Peskin (2002).

# THE STANDARD DISCRETE IB COUPLING

The numerical method replaces $\delta$ by a regularized kernel $\delta_h$ whose support is a few Eulerian grid cells wide:

$$\boldsymbol{f}_h = S_h \boldsymbol{F}_h, \qquad \boldsymbol{U}_h = J_h \boldsymbol{u}_h.$$

- $S_h$ spreads Lagrangian forces to nearby grid points.
- $J_h$ interpolates grid velocity to Lagrangian interaction points.
- Kernel support and moment conditions control smoothness, localization, and grid dependence.

Choose $J_h$ as the weighted discrete adjoint of $S_h$ so that the transferred power is consistent:

$$\langle \boldsymbol{u}_h, S_h \boldsymbol{F}_h \rangle_E = \langle J_h \boldsymbol{u}_h, \boldsymbol{F}_h \rangle_L .$$

> Regularization spreads a singular force layer and associated jump fields over $O(h)$. The Lagrangian boundary itself remains explicitly represented.

# WHY ADJOINT OPERATORS TRANSFER POWER CONSISTENTLY

With Eulerian grid spacing $h$ and Lagrangian quadrature weights $\Delta q_l$, define

$$\langle \boldsymbol{u}_h, \boldsymbol{f}_h \rangle_E = \sum_{\boldsymbol{x}} \boldsymbol{u}_h(\boldsymbol{x}) \cdot \boldsymbol{f}_h(\boldsymbol{x})\, h^3, \qquad \langle \boldsymbol{U}_h, \boldsymbol{F}_h \rangle_L = \sum_{l} \boldsymbol{U}_l \cdot \boldsymbol{F}_l\, \Delta q_l.$$

- Power the spread force does on the fluid: $P_E = \langle \boldsymbol{u}_h, S_h \boldsymbol{F}_h \rangle_E$.
- Power the structural force does at the interpolated structure velocity: $P_L = \langle J_h \boldsymbol{u}_h, \boldsymbol{F}_h \rangle_L$.
- If $J_h = S_h^*$, then $P_E = P_L$ for every $\boldsymbol{u}_h$ and $\boldsymbol{F}_h$: the transfer step neither creates nor destroys energy.

Here $\langle \cdot, \cdot \rangle_E$ and $\langle \cdot, \cdot \rangle_L$ are the discrete Eulerian and Lagrangian inner products induced by the quadrature rules. See Griffith and Patankar (2020), §3.3, who write these operators as $S_{\mathrm{IB}}$ and $J_{\mathrm{IB}}$ with $J_{\mathrm{IB}} = S_{\mathrm{IB}}^* = h^3 M^{-1} S_{\mathrm{IB}}^{\mathsf{T}}$, where $M$ is the diagonal matrix of Lagrangian quadrature weights (for uniform weights $\Delta q$, $J_{\mathrm{IB}} = (h^3/\Delta q)\, S_{\mathrm{IB}}^{\mathsf{T}}$).

# WHAT REGULARIZED IB COUPLING BUYS AND COSTS

**Strengths**

- No body-conforming fluid mesh or fluid-grid remeshing
- Large structural motion and deformation
- Thin immersed boundaries and transient contact
- Efficient Cartesian-grid solvers and adaptive mesh refinement
- Fiber, surface, and volumetric structural models

**Numerical challenges**

- Pressure and traction jumps are smoothed at finite $h$
- Interfacial forces and stresses may converge slowly
- Coarse interaction-point spacing can cause leakage or volume error
- Stiff elastic forces can severely restrict explicit time steps

AMR targets spatial resolution. Implicit or semi-implicit coupling targets stiffness. Sharp-interface corrections target jump accuracy. Finite-element structure models broaden constitutive and geometric choices.

# ACCURACY AND REGULARIZED BOUNDARY COUPLING

At finite grid spacing, $\delta_h$ averages information across a stencil several cells wide.

- The interpolated velocity need not equal the pointwise trace of a sharply resolved field.
- Pressure and stress jumps are represented over an $O(h)$ layer.
- Kernel moments, support, and smoothness influence conservation and grid dependence.

> Accuracy depends jointly on
>
> $h$, $\Delta s$ or quadrature density, $\Delta t$,
>
> the fluid solver, the structural discretization, and solution regularity.

A refinement study must vary every resolution that controls the coupled approximation.

# COMMON FAILURE MODES AND PRACTICAL CHECKS

**Interaction resolution**

- Interaction points too far apart can leak.
- A very fine structural mesh may add stiffness or poor conditioning without improving the fluid resolution.
- Different spreading and interpolation kernels break the usual adjoint power identity.

**Time integration and verification**

- Explicit elastic forces can impose severe time-step limits.
- Monitor volume, deformation, net force, and transferred power separately.
- Refine Eulerian, Lagrangian, quadrature, and temporal resolutions systematically.

> Many apparent “IB failures” are mismatches among the grid, interaction points, kernel, and time integrator rather than failures of the continuum formulation.

With nodal quadrature, a structure leaks when its Lagrangian nodes become too far apart relative to $h$ (Peskin 2002). Non-nodal (e.g. Gaussian) quadrature lets the structural mesh be coarse provided the net of quadrature points is dense enough (Griffith and Luo 2017; Griffith and Patankar 2020, §3.3).

# WHEN IS CLASSICAL IB A GOOD CHOICE?

**Strong fit**

- Large or complicated structural motion
- Thin, flexible, or actively deforming structures
- Transient contact without a body-fitted fluid mesh
- Quantities of interest that tolerate regularized interfacial fields

**Requires special care**

- Pointwise tractions or high-order interface stresses
- Thin high-Reynolds-number boundary layers
- Nearly rigid constraints or severe stiffness
- Tearing, merging, or reconnection, which require an explicit topology model

> High Reynolds number is not a formal exclusion. Published immersed FSI examples reach $\mathrm{Re} \approx 2 \times 10^4$, but thin shear layers and turbulence still require adequate resolution and modeling.

Griffith and Patankar, *Annu. Rev. Fluid Mech.* 52 (2020); the $\mathrm{Re} \approx 2 \times 10^4$ example is the bioprosthetic heart-valve application of their §5.3.1.

# DISCRETIZATION AND CONVERGENCE CHECKLIST

**Interaction geometry**

- Resolve the structure with enough Lagrangian interaction points or quadrature points relative to $h$.
- Coarse nodal coupling can leak. Excessively fine structural meshes add cost and may worsen stiffness or conditioning.
- Preserve weighted adjointness between spreading and interpolation.

**Verification**

- Refine $h$, the Lagrangian discretization or quadrature, and $\Delta t$ consistently.
- Check velocity, deformation, volume, loads, and energy transfer separately.
- Use an implicit or semi-implicit treatment when elastic stiffness controls stability.

> If the support of $\delta_h$ scales with $h$ and the coupled scheme is stable and consistent, refinement recovers the singular IB formulation. Convergence rates depend on solution regularity; standard regularized methods are often only first order near stress jumps.

Peskin (2002); Griffith and Patankar (2020); Griffith and Luo (2017).

# BODY-FITTED FSI: WHY CONFORMING MESHES ARE ATTRACTIVE

A body-fitted fluid mesh conforms to the moving interface.

- Boundary conditions and traction continuity can be imposed on the represented surface.
- Boundary layers and interfacial stresses can be resolved without regularized smearing.
- ALE, moving-mesh finite element, and moving-mesh finite volume schemes are common examples.

**Geometric bargain**

sharp, conforming interface



fluid-mesh motion and quality control

# BODY-FITTED FSI: COST UNDER LARGE MOTION

**Typical numerical work**

- solve for mesh motion or smoothing;
- detect distortion and remesh when needed;
- transfer fields between successive meshes;
- maintain coupled fluid–solid interface data structures.

**Best fit**

- interface accuracy is central;
- motion remains compatible with mesh quality;
- the added geometric machinery is justified by the quantities of interest.

> Large deformation is possible in sophisticated body-fitted schemes; it is the mesh-management burden, not a universal deformation cutoff, that distinguishes the tradeoff.

# COMPARISON WITH OTHER FSI APPROACHES

| Method family | Fluid and interface treatment | Typical numerical challenge |
|---|---|---|
| Regularized IB | Fixed, nonconforming grid; $\delta_h$ spreading and interpolation | Smeared jump fields, interfacial accuracy, leakage, and stiffness |
| ALE / body-fitted | Moving, boundary-conforming fluid mesh | Mesh quality, mesh motion, and possible remeshing under large motion |
| Immersed interface / sharp immersed | Fixed grid; interface-aware jump corrections | Geometric reconstruction and jump-dependent stencils or solvers |
| Fictitious domain / penalization | Fixed background grid; distributed constraints or penalties | Conditioning, penalty or modeling error, and method-specific coupling |

> The deformation a method can handle depends on its interface representation, mesh strategy, solver, and implementation. Small, moderate, and large are not reliable method-family labels.

# THE IMMERSED INTERFACE METHOD

The immersed interface method (IIM) keeps a fixed Eulerian grid and incorporates analytically derived jump conditions into nearby stencils.

- It avoids regularized smearing of jump fields.
- It can improve near-interface velocity, pressure, deformation, and traction accuracy.
- It adds geometry reconstruction and jump-dependent stencils or solvers.

Define $[\![a]\!] = a^+ - a^-$ and take $\boldsymbol{n}$ from $\Omega^-$ to $\Omega^+$. If $\boldsymbol{f}_\Gamma$ is the interface-on-fluid force in $+\boldsymbol{f}_\Gamma \delta_\Gamma$, then

$$[\![\boldsymbol{u}]\!] = 0, \qquad [\![\boldsymbol{\sigma}]\!]\,\boldsymbol{n} = -\boldsymbol{f}_\Gamma.$$

Traction balance determines jumps in pressure and velocity derivatives. Pressure may be continuous or discontinuous, depending on the interfacial force.

> Sharp means resolving jumps without regularized smearing; it does not mean the computed solution is exact. Large or topology-changing motion still depends on the interface and geometry algorithms.

Li and Lai, *J. Comput. Phys.* 171 (2001); Griffith and Patankar (2020).

# WORKED STEP: THE SIGN OF THE STRESS JUMP

Consider a steady one-dimensional balance with a point force at an interface $x = 0$, normal $\boldsymbol{n} = +\boldsymbol{e}_x$ pointing from $\Omega^- = \{x < 0\}$ to $\Omega^+ = \{x > 0\}$:

$$0 = \frac{\partial \sigma}{\partial x} + f_\Gamma \, \delta(x).$$

Integrate across $(-\epsilon, \epsilon)$ and let $\epsilon \to 0$:

$$\sigma(0^+) - \sigma(0^-) + f_\Gamma = 0 \qquad \Longrightarrow \qquad \boxed{[\![\sigma]\!] = -f_\Gamma}.$$

In several dimensions the same pillbox argument gives $[\![\boldsymbol{\sigma}]\!]\,\boldsymbol{n} = -\boldsymbol{f}_\Gamma$. With $\boldsymbol{\sigma} = -p\boldsymbol{I} + \mu(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^{\mathsf{T}})$, the normal component yields the pressure jump and the tangential components yield jumps in velocity derivatives.

> If the force pushes the fluid toward $\Omega^+$ ($f_\Gamma > 0$), the stress on the $+$ side is lower; for a fluid at rest ($\sigma = -p$) this means $p(0^+) = p(0^-) + f_\Gamma$, so the force compresses the fluid ahead of it.

Pillbox derivation in the spirit of Peskin (2002) and Li and Lai (2001).

# IMMERSED INTERFACE METHODS: STRENGTHS AND TRADEOFFS

**Strengths**

- fixed background grid;
- sharp representation of pressure and stress jumps;
- potentially higher near-interface accuracy;
- no regularized force layer in the target continuum description.

**Added requirements**

- interface geometry and normals;
- jump conditions consistent with the physical model;
- corrected stencils or specialized solvers;
- robust handling of large motion and topology changes.

> IIM prioritizes sharp jump accuracy; regularized IB prioritizes a simple, robust transfer mechanism. The right choice follows the observable and geometry.

# OPEN-SOURCE SOFTWARE AND TUTORIALS

## IBAMR

- Distributed-memory C++ framework
- Cartesian adaptive mesh refinement
- Examples for classical IB, IBFE, constrained IB, and IIM in two and three dimensions

`ibamr.github.io`
Griffith et al., *J. Comput. Phys.* 223 (2007).

## IB2d

- MATLAB and Python implementations in two dimensions
- More than 75 examples for teaching and research
- Introductory video tutorials linked from the repository

`github.com/nickabattista/IB2d`
Battista, Strickland, and Miller, *Bioinspir. Biomim.* 12 (2017).

> Nick Battista's introduction to the IB method is the first tutorial linked by IB2d. Course-specific lectures by Matea Santiago may require Google Drive access.

# WHY THE OPEN-SOURCE ECOSYSTEM MATTERS

**For learning and verification**

- inspect the actual spreading and interpolation operators;
- reproduce benchmark problems;
- compare kernels, solvers, and time integrators;
- trace continuum notation into implementation.

**For research**

- prototype new constitutive or active-force models;
- use tested parallel solvers and AMR infrastructure;
- document algorithmic choices for reproducibility;
- separate model development from low-level solver development.

> Software availability is a practical advantage, not a mathematical accuracy claim. Validation is still problem- and implementation-specific.

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- Z. Li and M.-C. Lai, *The immersed interface method for the Navier–Stokes equations with singular forces*, *Journal of Computational Physics* 171 (2001), 822–842. doi:10.1006/jcph.2001.6813.


> Course copies of the first three papers are available in the repository's `papers/` directory.

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- N. A. Battista, W. C. Strickland, and L. A. Miller, *IB2d: a Python and MATLAB implementation of the immersed boundary method*, *Bioinspiration & Biomimetics* 12 (2017), 036003. doi:10.1088/1748-3190/aa5e08.
- Wikimedia Commons, figure “ImmersedBoundary grid,” used in the Wikipedia article “Immersed boundary method,” https://en.wikipedia.org/wiki/Immersed_boundary_method. Source of the schematics `ImmersedBoundary_grid.png` and `ImmersedBoundary_grid2.png`; licence to be confirmed.